

Jonathan Villanueva

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EDUCATION

University of the Pacific — Bachelor of Computer Science
Fall 2025

Relevant Coursework: Data Structures & Algorithms, Computer Theory, Database Systems, Data Analytics, System Administration & Security, Application Development, computer networks and the internet

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Core Competencies: Algorithms & Data Structures, Software Development, Object-Oriented Programming, RESTful API, Debugging & Testing

Frameworks: React, Next.js, Express, FastAPI

Databases: PostgreSQL

Tools: Git, NetworkX, npm, Alembic, Docker

PROJECTS

Python Compiler | Python

- Designed and implemented a compiler capable of performing lexical analysis, parsing, semantic validation, and code generation.
- Developed modular compiler components to translate source code into executable representations.
- Debugged and optimized compiler functionality through iterative testing and validation.

Network Throughput Router Simulator | Python, NetworkX

- Engineered a routing simulator that maximizes network bandwidth using a modified Dijkstra algorithm prioritizing throughput over shortest path.
- Modeled complex network topologies and visualized routing behavior to analyze bottlenecks and performance tradeoffs.
- Evaluated algorithm performance through simulation and data-driven analysis.

WatchBuddy | Next.js, TypeScript, React, PostgreSQL

- Built a full-stack social streaming platform integrating external APIs and persistent user data.
- Designed scalable frontend architecture using reusable React components and optimized application routing.
- Implemented RESTful APIs for authentication, user relationships, watch parties, and data management.
- Collaborated with AI tools throughout development to accelerate prototyping and debugging.